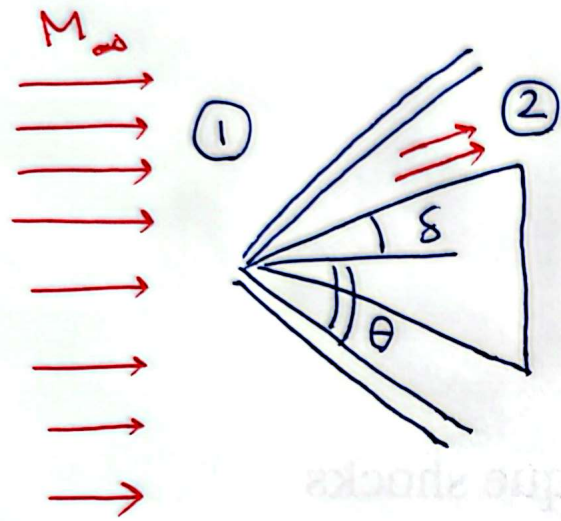
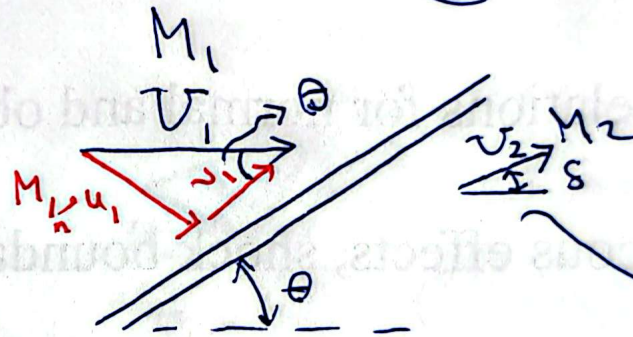




oblique shock rotates the flow
s. that flow becomes parallel
to wedge.

$$U_2 < U_1$$

$$M_2 < M_1$$



$\theta \equiv \beta$: shock angle

$\delta \equiv \theta$: wedge angle

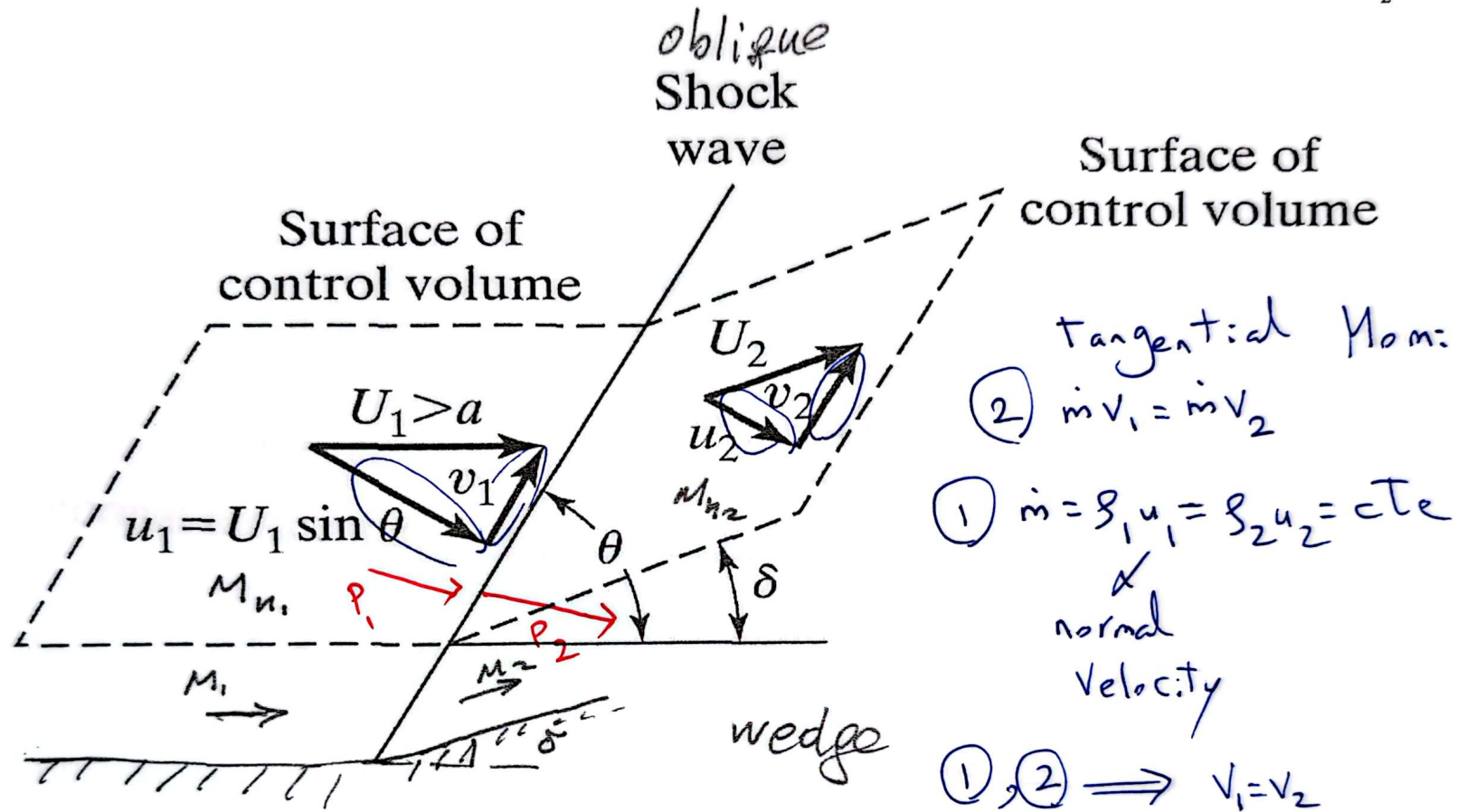
(flow deflection
angle)

$$\begin{cases} u_1 = U_1 \sin \theta \\ M_{1n} = M_1 \sin \theta \end{cases}$$

normal
Mach

$$\begin{cases} M_{1t} = M_1 \cos \theta \\ v_1 = U_1 \cos \theta \end{cases}$$

$$\begin{cases} u_2 = U_2 \sin(\theta - \delta) \\ M_{2n} = M_2 \sin(\theta - \delta) \\ M_{2t} = M_2 \cos(\theta - \delta) \\ v_2 = U_2 \cos(\theta - \delta) \end{cases}$$



Consider control volume around the shock

let $u_1, u_2 \Rightarrow$ velocity components $\perp$ the shock

$v_1, v_2 \Rightarrow$ velocity components $\parallel$ to the shock

continuity

$$\rho_1 u_1 = \rho_2 u_2$$

(8.64)

tangential momentum

$$\rho_1 u_1 v_1 = \rho_2 u_2 v_2$$

(8.66)

(same tangential components)

$\Rightarrow$

$$v_1 = v_2$$

(8.70)

normal momentum:

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

has pressure

(8.65)

energy equation (for adiabatic flow)

$$h_1 + \frac{U_1^2}{2} = h_2 + \frac{U_2^2}{2}$$

(8.67)

$$(U^2 = u^2 + v^2)$$

$$\vec{U} = u\vec{x} + v\vec{y}$$

As $V_1 = V_2$ energy Eq. simplifies to: 4

$$\Rightarrow \boxed{h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}} \quad (8.71)$$

Also, perfect gas $p = \rho RT$ (1.10)

Enthalpy equation: $h = c_p T$ (8.15)

Normal component of velocity is used above:

$$\underbrace{u_1}_{\text{Normal velocity}} = U_1 \underbrace{\sin \theta}_{\text{shock comp. angle}} \Rightarrow M_{n_1} = M_1 \sin \theta \quad (8.72)$$

System of 5 eqs with 5 unknowns can be solved to find qualities behind the shock (using M_{n1})

$$M_1 \geq 5, \quad \delta \approx 1^\circ, \quad \theta \approx 19.37^\circ$$

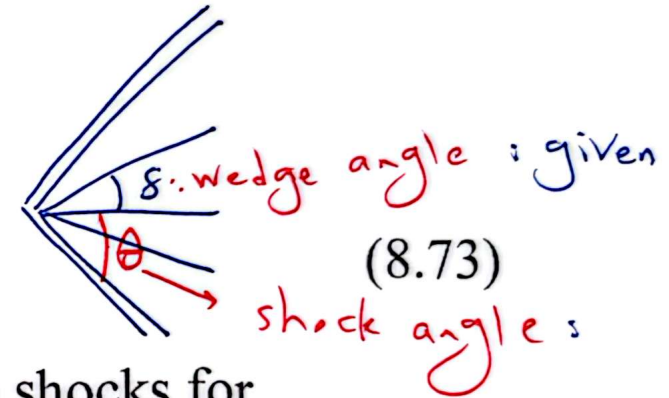
5

online calc.

M- δ - θ relation (need 2 out of 3)

$$\cot \delta = \tan \theta \left[\frac{(\gamma + 1)M_1^2}{2(M_{n1}^2 - 1)} - 1 \right]$$

known δ (pointing to $\cot \delta$)
 unknown θ (pointing to $\tan \theta$)
 known M_1 (pointing to M_1^2)
 known M_{n1} (pointing to M_{n1}^2)

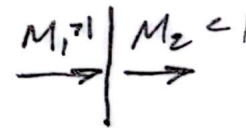


(8.73)

- normal shocks are a special case of oblique shocks for

$$\theta = \frac{\pi}{2}, \quad \delta = 0$$

(no deflection)



$$\frac{p_2}{p_1} = \frac{2\gamma M_{n1}^2 - (\gamma - 1)}{\gamma + 1} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n1}^2 - 1)$$

(8.74)

Accurate
 for normal
 and oblique
 shocks

Normal shock
 put $\theta = \frac{\pi}{2}$

$$M_{n1} = M_1 \sin \theta = M_1$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_{n1}^2}{(\gamma - 1)M_{n1}^2 + 2}$$

(8.75)

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \frac{\rho_1}{\rho_2} \Rightarrow \frac{T_2}{T_1} = \frac{[2\gamma M_{n_1}^2 - (\gamma - 1)][(\gamma - 1)M_{n_1}^2 + 2]}{(\gamma + 1)^2 M_{n_1}^2} \quad (8.76)$$

$$M_{n_2}^2 = \frac{M_{n_1}^2 + [2 / (\gamma - 1)]}{[2\gamma / (\gamma - 1)] M_{n_1}^2 - 1} = \frac{1 + [(\gamma - 1) / 2] M_{n_1}^2}{\gamma M_{n_1}^2 - (\gamma - 1) / 2}$$

The Mach number behind the shock can be found from

$$\boxed{M_2 = \frac{M_{n_2}}{\sin(\theta - \delta)}} \Rightarrow M_2^2 = \frac{1 + [(\gamma - 1) / 2] M_{n_1}^2}{[\gamma M_{n_1}^2 - (\gamma - 1) / 2] \sin^2(\theta - \delta)}$$

$$\frac{p_{t_2}}{p_{t_1}} = e^{-(s_2 - s_1)/R} = \left[\frac{(\gamma + 1) M_{n_1}^2}{(\gamma - 1) M_{n_1}^2 + 2} \right]^{\gamma/(\gamma - 1)} \left[\frac{(\gamma + 1)}{2\gamma M_{n_1}^2 - (\gamma - 1)} \right]^{1/(\gamma - 1)} \quad (8.78)$$

$$p_{t_2} < p_{t_1}$$

Please note the change in entropy

$$s_2 - s_1 < 0 \quad \text{for} \quad M_{n_1} < 1$$

entropy cannot reduce
(not physical)

$$s_2 - s_1 = 0 \quad \text{for} \quad M_{n_1} = 1 \quad (\text{sound/Mach wave})$$

$$s_2 - s_1 > 0 \quad \text{for} \quad M_{n_1} > 1 \quad (\text{shock wave})$$

$$C_p = \frac{p_2 - p_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_1^2}$$

Trends and graphs are below

$M_1 < 1 \rightarrow$ No shock
as $\Delta S < 0$
"non-physical"

$M = 1$ } $\Delta S = 0$
sound wave

$M > 1$ || $\Delta S > 0$
shock wave

TABLE 8.4 Flow Property Changes across Weak Shock Waves

Normal shock ① ②	Oblique shock ① // ②
$p_2 > p_1$ $\rho_2 > \rho_1$ $T_2 > T_1$ $M_2 < 1$ $p_{t,2} < p_{t,1}$ $T_{t,2} = T_{t,1}$	$p_2 > p_1$ $\rho_2 > \rho_1$ $T_2 > T_1$ $M_2 < M_1$ $p_{t,2} < p_{t,1}$ $T_{t,2} = T_{t,1}$
$M_1 > 1$ $M_2 < 1$	$M_1 > 1$ // $M_2 < M_1$ $M_2 > 1$ or $M_2 < 1$ Weak shock or strong shock total pressure decreases (adiabatic)

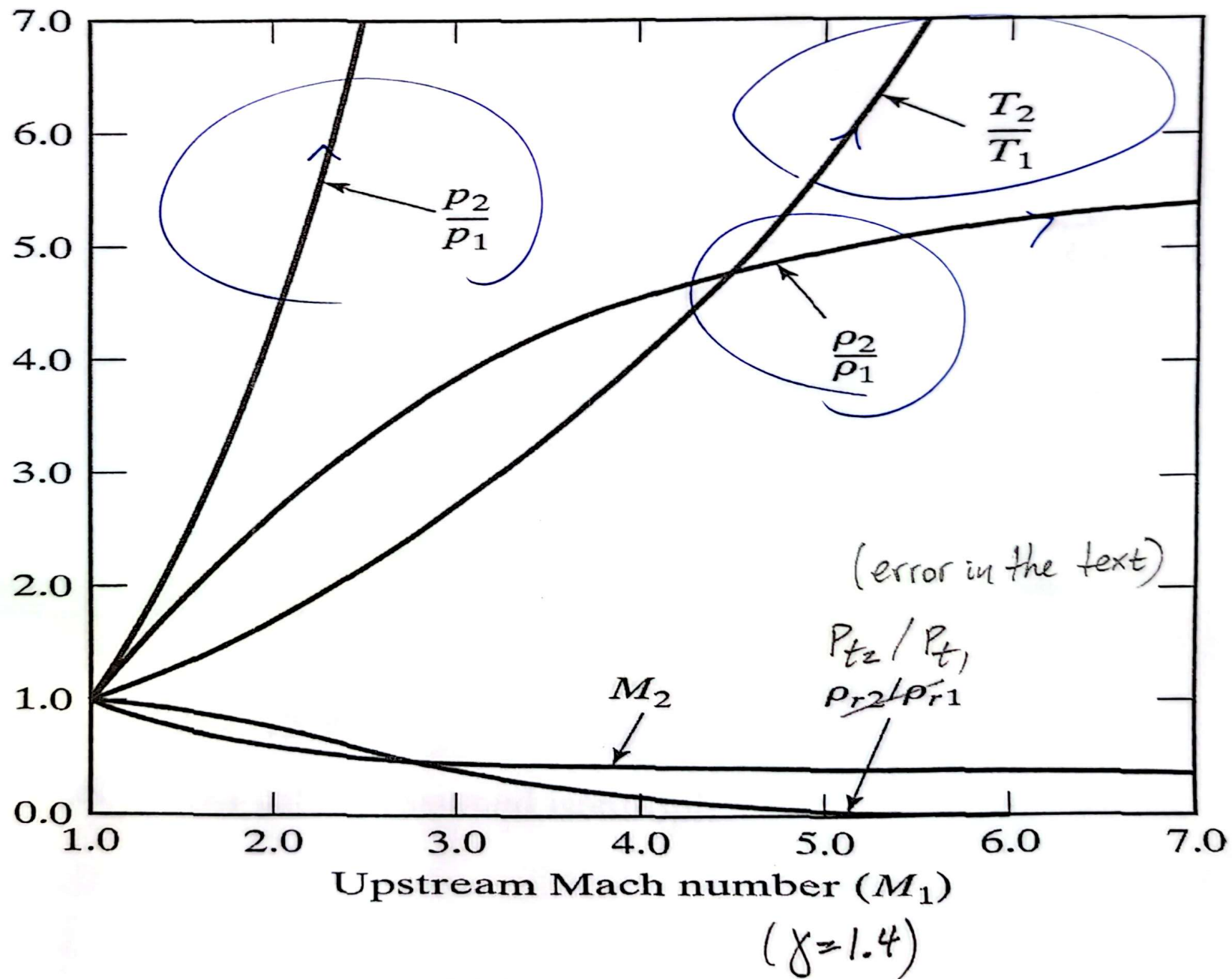
limits

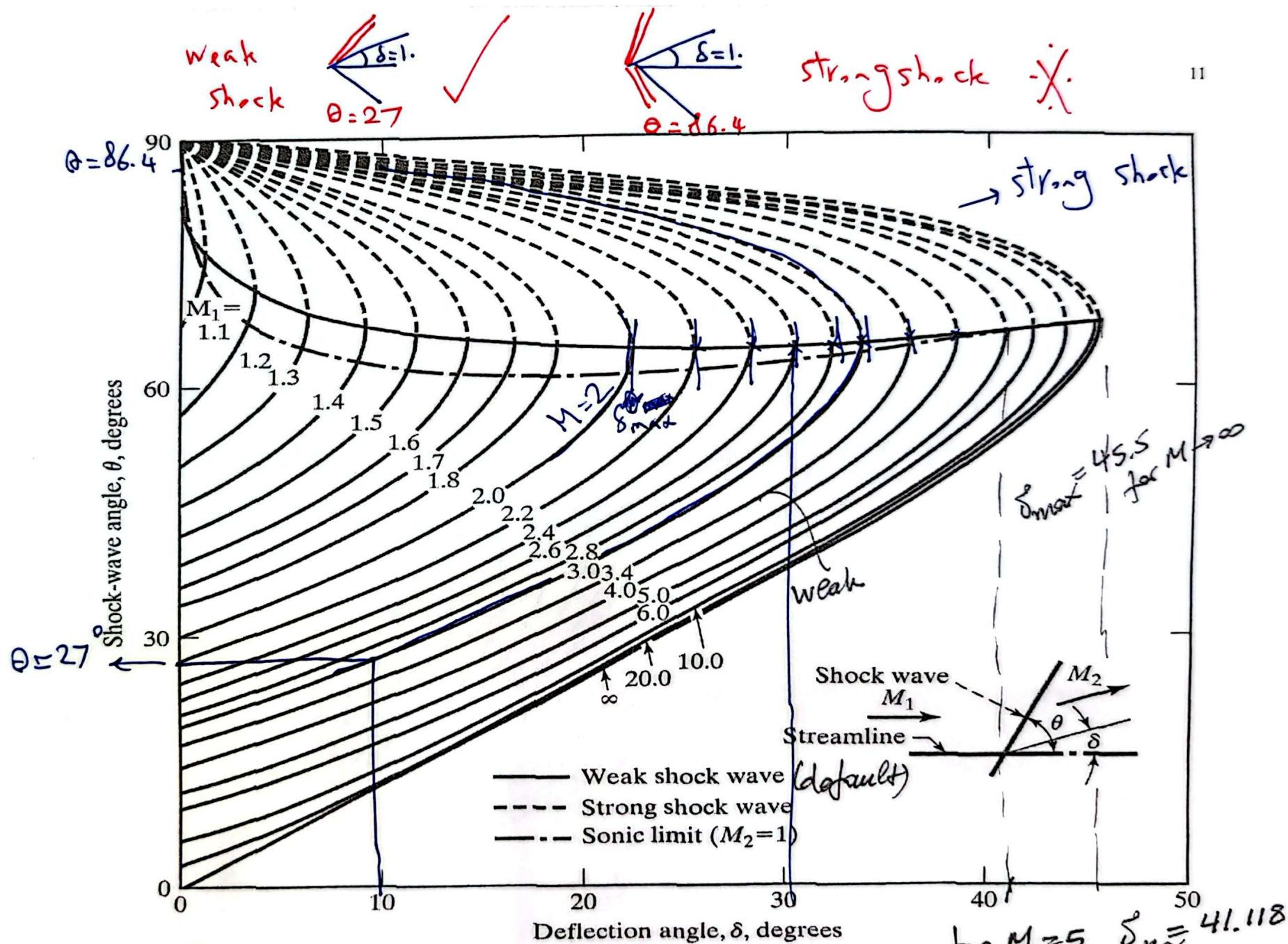
for $M_1 \rightarrow \infty$ $M_2 \rightarrow \sqrt{\frac{\gamma-1}{2\gamma}} = 0.378$

$$\frac{p_2}{p_1} \Rightarrow \frac{\gamma+1}{\gamma-1} = 6$$

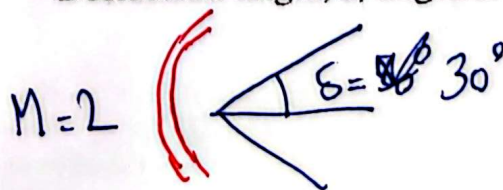
(strength) $\frac{p_2}{p_1} \rightarrow \infty$

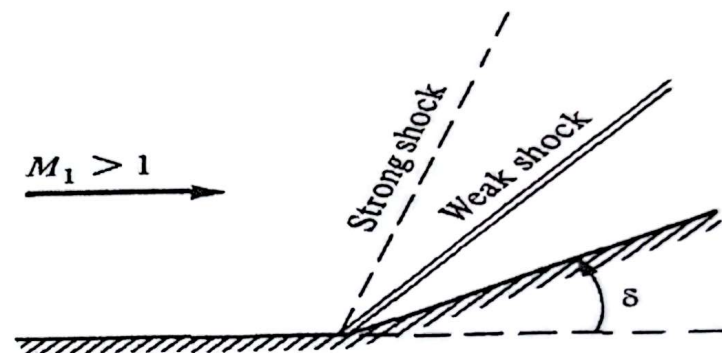
$$\frac{T_2}{T_1} \rightarrow \infty$$



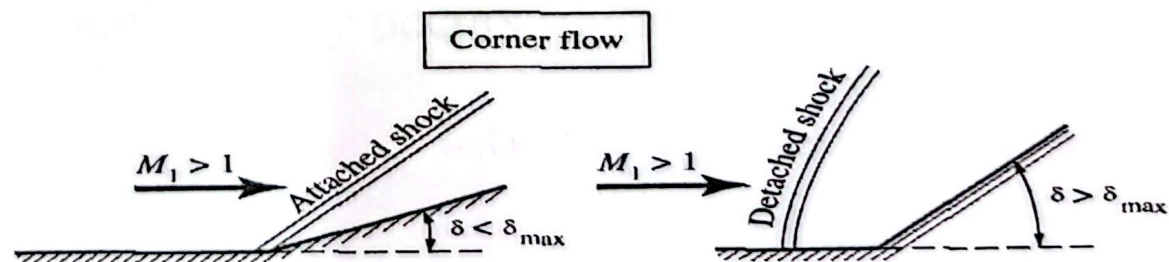
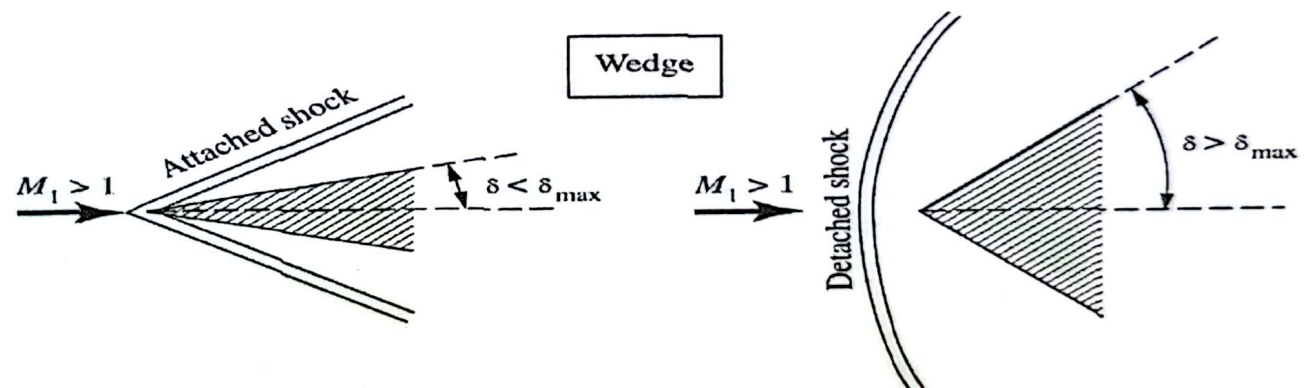


If for given Mach
 $\theta > \theta_{max}$, shock detaches





strong shock
has a higher θ



- For any given M_1 there is a $\delta_{\max}$ (see graphs)

$\delta > \delta_{\max}$ detached shock

$\delta < \delta_{\max}$ 2 values of θ (shock angle)

- large (strong shock)
- small (weak shock)

Back pressure determines the solution;

weak shock usually occurs (default)

use unless specified otherwise

for a *strong* shock $M_2 < 1$

for a *weak* shock $M_2 > 1$ (except near $\theta_{\max}$)

$$\delta = 0 \Rightarrow \theta = \frac{\pi}{2} \quad (\text{strong}) \quad \text{normal shock}$$

$$\rightarrow \theta = \mu \quad (\text{weak}) \quad \text{Mach wave}$$

- for a fixed δ as $M_1 \downarrow$ then $\theta \uparrow$ till the shock detaches.
(weak solution)
- as $M_{n_1} \uparrow \Rightarrow$ stronger shock $\frac{P_2}{P_1} \uparrow$
- as $\delta \uparrow$ (with M_1 constant) $\Rightarrow$ stronger shock, $\theta \uparrow$ (weak)
- as $M_1 \uparrow$ (with δ constant) $\Rightarrow$ stronger shock, $\theta \downarrow$ (weak)

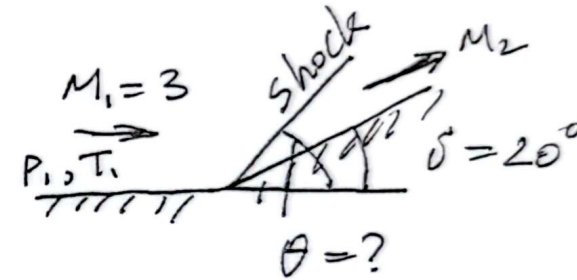
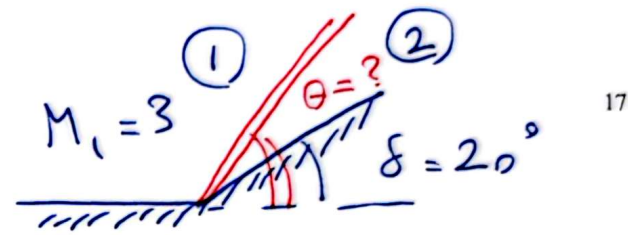
example

online calc

$$p_2 = \frac{p_2}{p_1} \times p_1 \rightarrow \text{given}$$
$$T_2 = \frac{T_2}{T_1} \times T_1 \rightarrow \text{given}$$

$$M_1 = 3, p_1 = 1 \text{ atm}, T_1 = 288 \text{ K}, \delta = 20^\circ$$

find $p_2, T_2, M_2, p_{t2}, T_{t2}$



from $\theta - \delta - M$ diagram: $M_1 = 3, \delta = 20^\circ \Rightarrow \theta = 37.5^\circ$ (weak)

$$\Rightarrow M_{n1} = M_1 \sin \theta = 1.826 \quad (>1) \text{ If don't want using}$$

from normal shock tables, for $M_{n1} = 1.826$

$$M_{n1} = 1.826 \Rightarrow \frac{p_2}{p_1} = 3.723 \quad ; \quad \frac{T_2}{T_1} = 1.551 \quad \text{go to}$$

Normal shock table and read data

$$M_{n_2} = 0.6108$$

$$p_{t_2}/p_{t_1} = 0.8011$$

$$\Rightarrow p_2 = 3.723 \text{ atm}, \quad T_2 = 446.7^\circ\text{K}, \quad M_2 = \frac{M_{n_2}}{\sin(\theta - \delta)} = 2.03 \quad (>1)$$

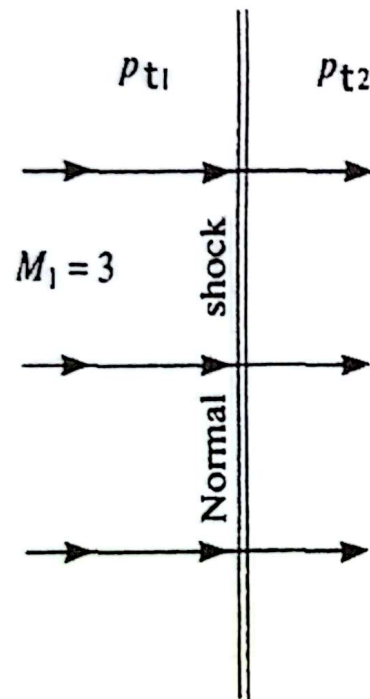
$M_2 < M_1$

from isentropic tables

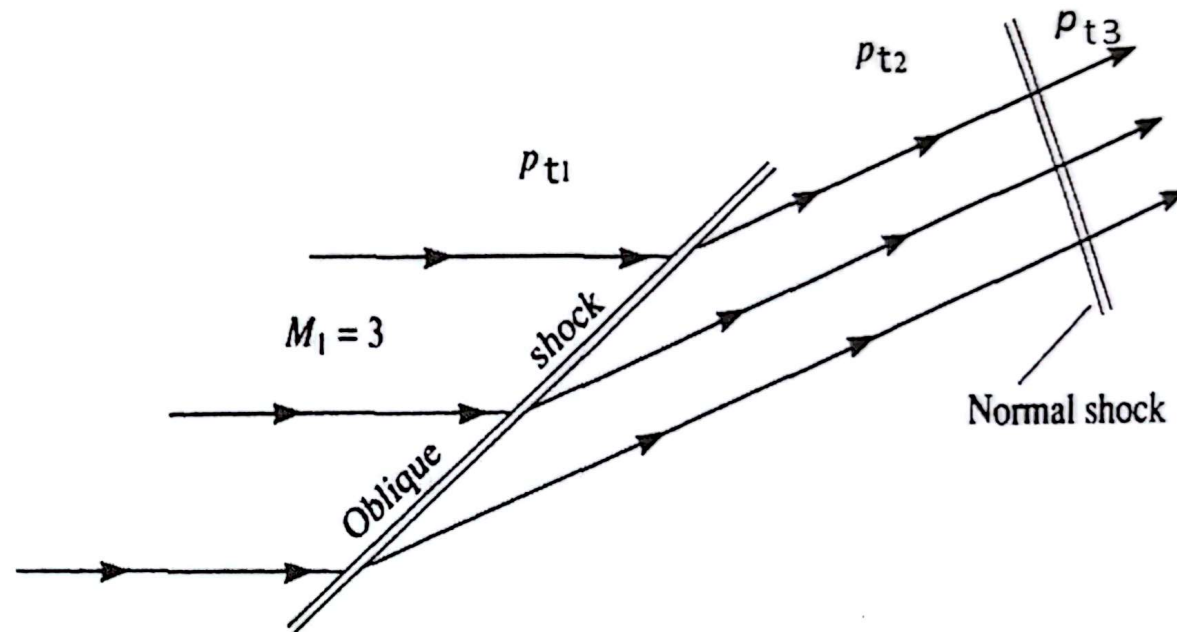
$$M_1 = 3 \Rightarrow \frac{p_{t1}}{p_1} = 36.73, \quad \frac{T_{t1}}{T_1} = 2.8$$

$$\underbrace{T_{t2} = T_{t1}}_{\text{adiabatic}} = \frac{T_{t1}}{T_1} T_1 = 806.4$$

$$p_{t2} = \underbrace{\frac{p_{t2}}{p_{t1}}}_{0.8011} \cdot \underbrace{\frac{p_{t1}}{p_1}}_{36.73} p_1 = 29.42 \text{ atm}$$



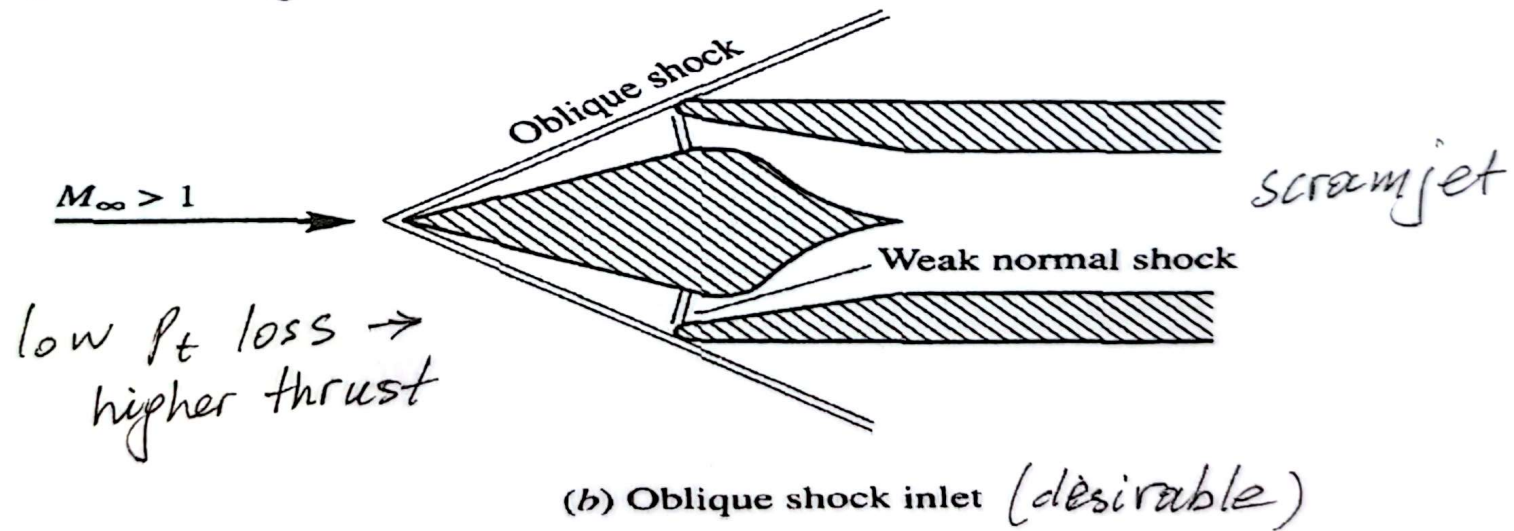
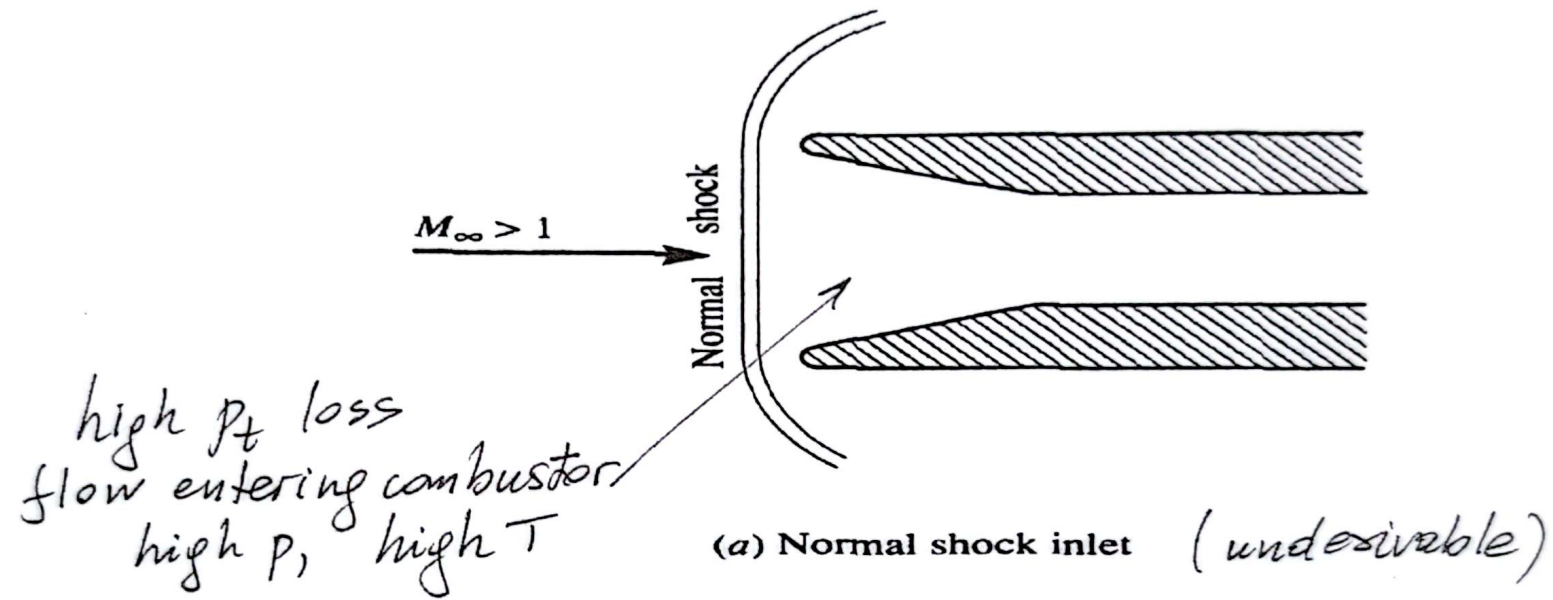
Case 1



Case 2

Final total pressure is 76% higher for case 2 (for $\theta = 40^\circ$)

[do it as an exercise]



$$C_p = \frac{p - p_\infty}{q_\infty}$$

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 = \frac{1}{2} \frac{\gamma p_\infty}{\gamma p_\infty} \rho_\infty U_\infty^2 = \frac{\gamma p_\infty U_\infty^2}{2 a_\infty^2}$$

$$= \frac{\gamma p_\infty U_\infty^2}{2 a_\infty^2} = \frac{\gamma}{2} p_\infty M_\infty^2$$

$$\Rightarrow C_p = \frac{p - p_\infty}{\frac{\gamma}{2} p_\infty M_\infty^2} \Rightarrow$$

$$C_p = \frac{2}{\gamma M_\infty^2} \left(\frac{p}{p_\infty} - 1 \right)$$

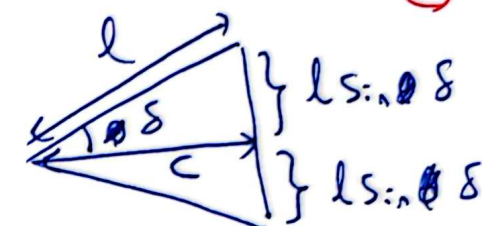
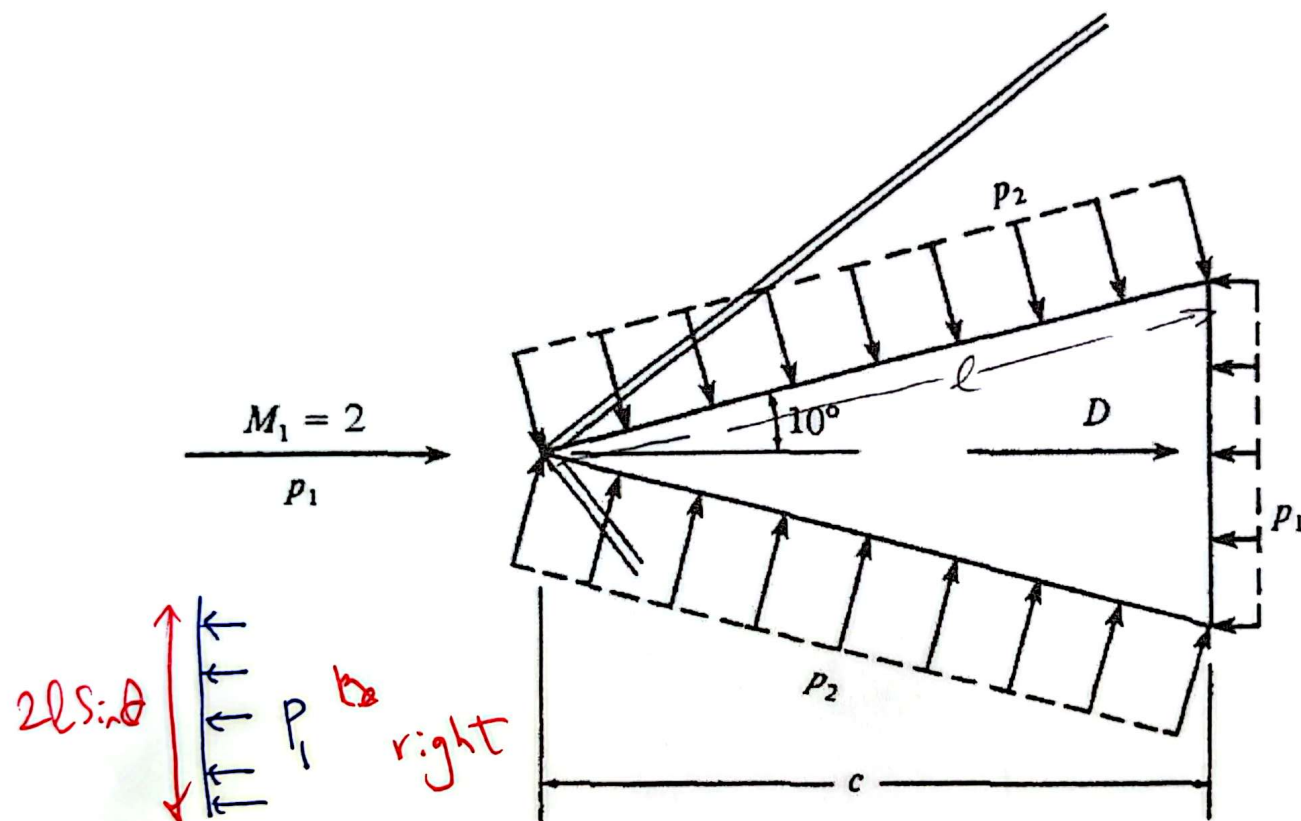
for [∞]1 and 2 ^{nothing} conditions:

$$C_p = \frac{2}{\gamma M_1^2} \left(\frac{p_2}{p_1} - 1 \right)$$

for incompressible ³⁰⁷flow:

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 \quad \left(\text{using Bernoulli's eq.} \right)$$

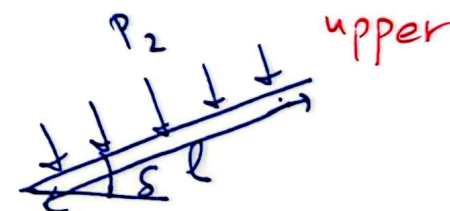
Consider a 10° wedge, $M_1=2$ find C_p , C_d



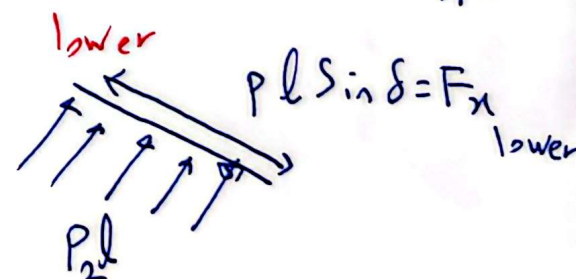
$$l \cos \delta = c$$

$$\rightarrow l = \frac{c}{\cos \delta}$$

$$l \sin \delta = c \tan \delta$$



$$p l \sin \delta = F_{x \text{ upper}}$$

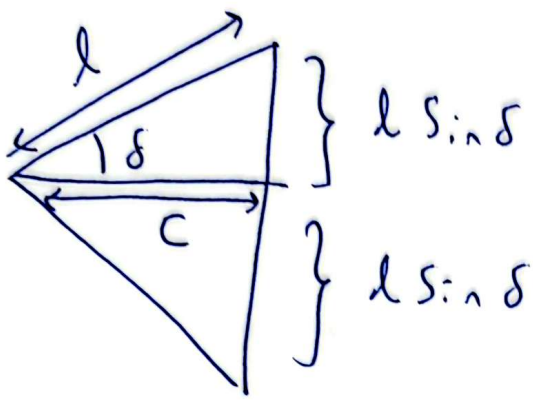


$$p l \sin \delta = F_{x \text{ lower}}$$

online calc.
 $M_1=2, \delta=10^\circ$ from $\theta-\delta$ M diagram $\theta = 39.314^\circ$

$$F_x = 2 p_2 l \sin \delta - 2 p_1 l \sin \delta$$

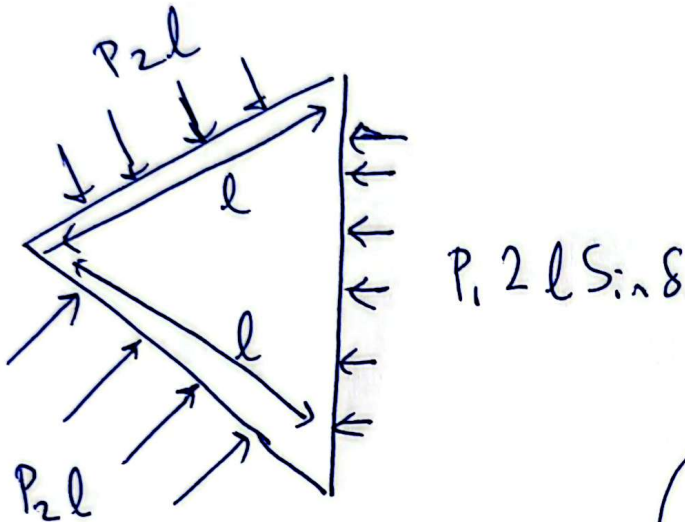
$$F_x = 2 (p_2 - p_1) c \tan \delta$$



$$l \cos \delta = c$$

$$l = \frac{c}{\cos \delta}$$

$$\Rightarrow l \sin \delta = c \tan \delta$$



$$F_{x \text{ upper}} = F_{x \text{ lower}} \\ = p_2 l \sin \delta$$

$$\Rightarrow F_x = 2 p_2 l \sin \delta$$

from inclined
surfaces

$$F_x = - 2 p_1 l \sin \delta$$

from vertical side

$$\Rightarrow F_x = 2 l \sin \delta (p_2 - p_1) =$$

$$2 c \tan \delta (p_2 - p_1) = D$$

(p-1)

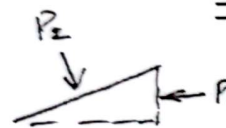
$$C_D = \frac{D}{\rho_{\infty} C} = \frac{D}{\underbrace{\frac{\gamma}{2} \rho_1 M_1^2}_{\rho_{\infty} = \rho_1} C} = \frac{\cancel{4} \cancel{\rho_1} \tan \delta (P_2 - P_1)}{\cancel{\frac{\gamma}{2}} \rho_1 M_1^2 \cancel{C}} = \left(\frac{4 \tan \delta}{\gamma M_1^2} \times \right. \quad \textcircled{1} \quad \left. \left(\frac{P_2}{P_1} - 1 \right) \right)$$

$$C_P = \frac{2}{\gamma M_1^2} \left(\frac{P_2}{P_1} - 1 \right) \quad \textcircled{2}$$

① and ② gives: $C_D = 2 \tan \delta C_P = 2 \tan 10^\circ \times 0.252$
 $= 0.184$

$$\Rightarrow M_{n_1} = M_1 \sin \theta = 2 \times \sin 39.314 = 1.267$$

from normal shock tables: $M_{n_1} = 1.267 \Rightarrow \frac{p_2}{p_1} = 1.707$



$$\Rightarrow C_p = \frac{p_2 - p_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_1^2} = \frac{4 \times (1.267^2 - 1)}{2.4 \times 2^2} = 0.252$$

also: $C_p = \frac{2}{\gamma M_1^2} \left(\frac{p_2}{p_1} - 1 \right) = 0.252$ same answer

$$D' = 2p_2 l \sin \delta - 2p_1 l \sin \delta = 2 \sin \delta l (p_2 - p_1)$$

/ 2 oblique sides vertical side
- horizontal component

drag / unit span $= 2 \tan \delta (p_2 - p_1)$ $l = c / \cos \delta$

- (or use gauge pressure)

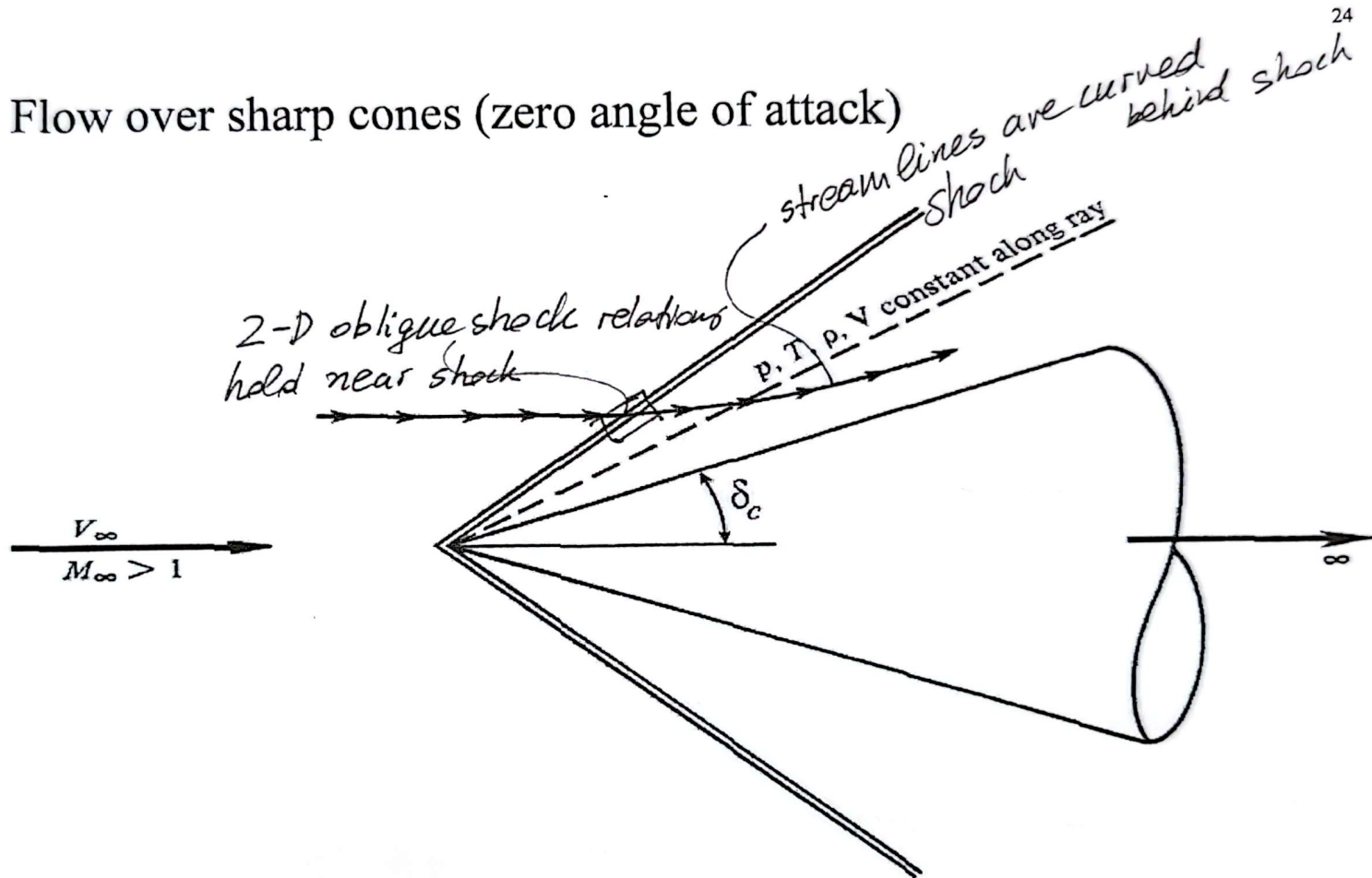
$$C_d = \frac{D'}{q_1 c} = \frac{D'}{\frac{\gamma}{2} p_1 M_1^2 c} = \frac{4 \tan \delta}{\gamma M_1^2} \left(\frac{p_2}{p_1} - 1 \right) = 2 \tan \delta C_p$$

$$= 2 \times \tan 10^\circ \times 0.252$$

$$= 0.184$$

note $C_d \neq 0$; oblique shock creates drag

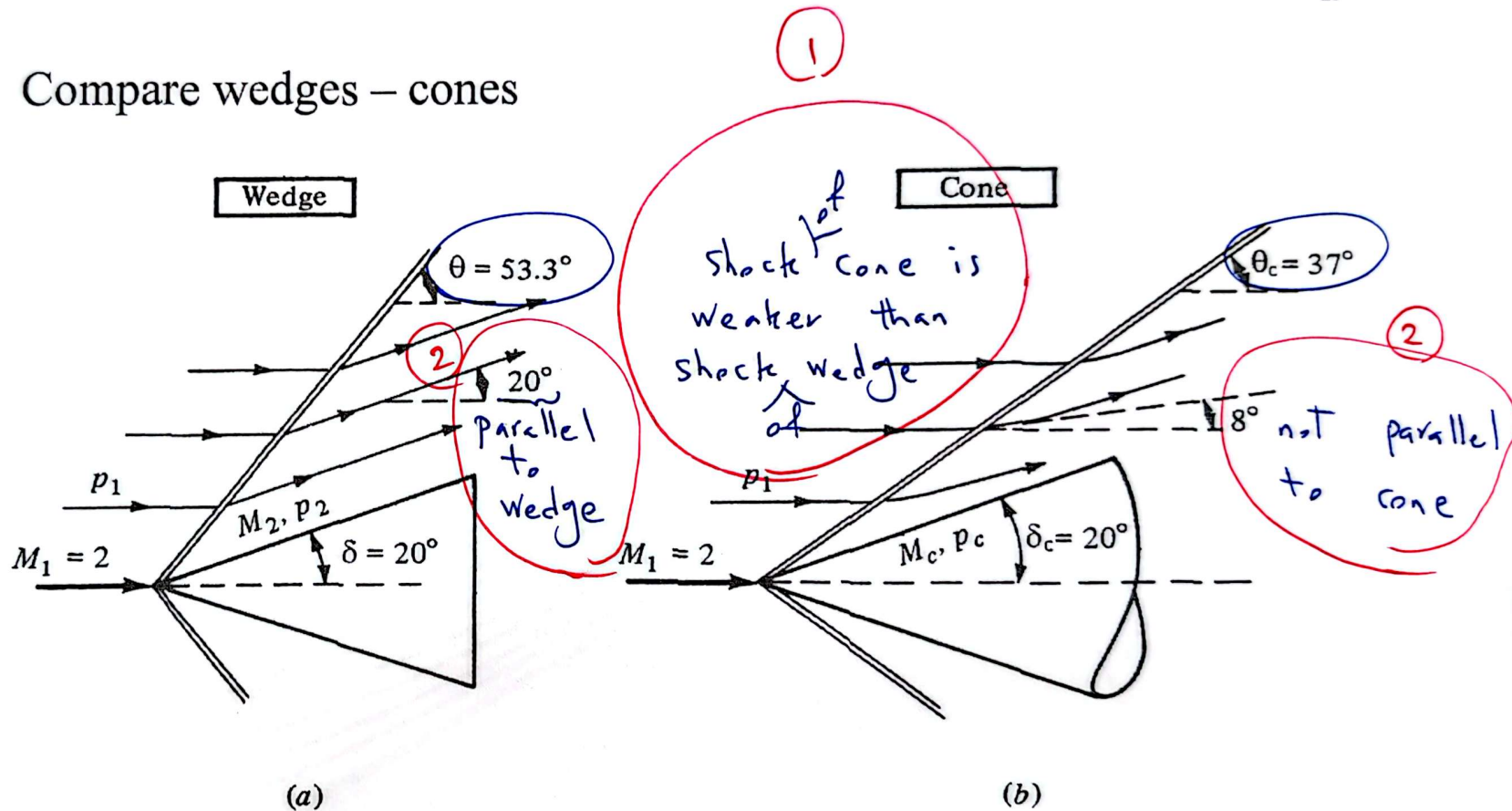
Flow over sharp cones (zero angle of attack)



$$C_p = \frac{p_c - p_1}{q_1}$$

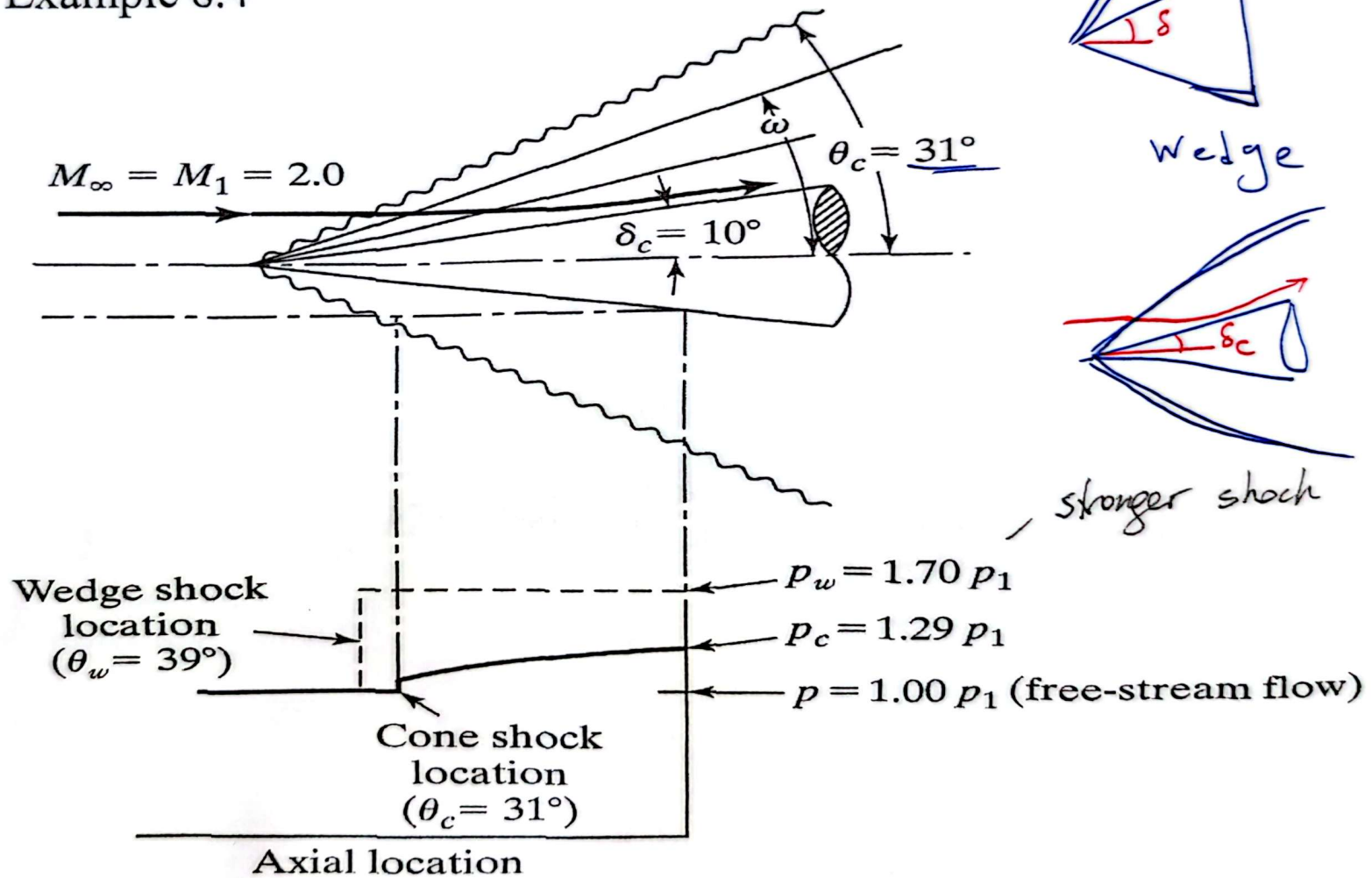
p_c : pressure on the cone surface
(constant along the surface)

Compare wedges – cones



θ is smaller for a cone; 3rd dimension gives extra space for the flow to move around object $\Rightarrow$ 3-D relieving effect
(similar to flow over cylinder / sphere)

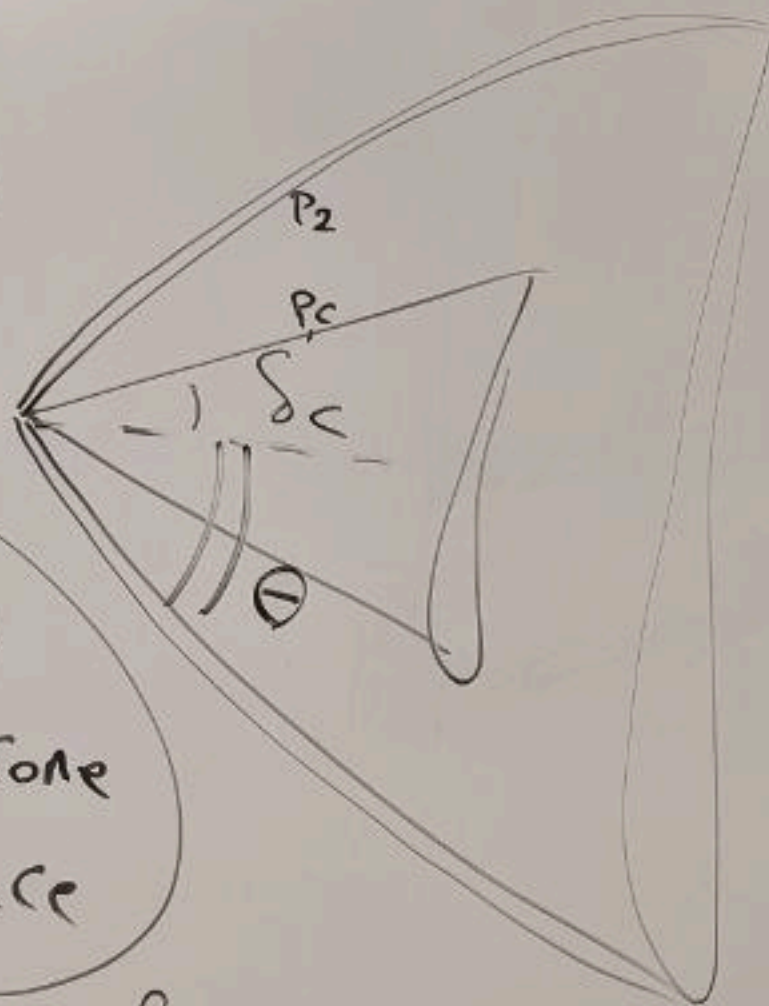
Example 8.4



used for drag force P_1

P_c : pressure on the cone surface

P_2 : pressure after



$$M_1 = 2$$

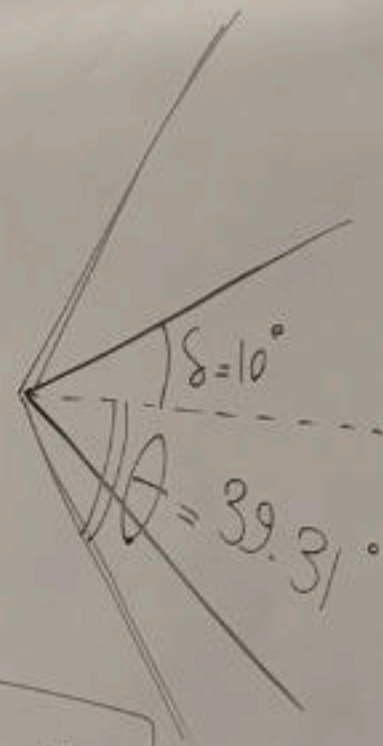
Online calc

$$\frac{P_2}{P_1} = 1.7$$

$$\frac{P_{o2}}{P_{o1}} = 0.97$$

C_p ?

C_d ?



Semi-vertex angle $\delta_c = 10^\circ$ $M_1 = 2$

^{online calc., cone part}
~~Fig 8.16a (cone graphs)~~ $\theta_c = 31^\circ$

Shock tables (2D) or (8.74) for $M_1 = 2$ $\theta_c = 31^\circ$ $p_2/p_1 = \underline{1.07}$ ^{1.088 ✓}

On the cone surface: $p_c/p_1 = 1.29$ ✓

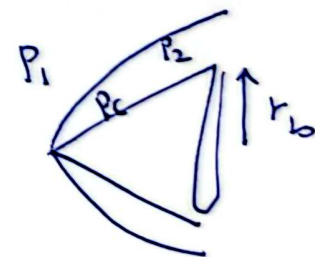
$$C_{p_c} = \frac{p_c - p_1}{q_1} = \frac{2}{\gamma M_1^2} \left(\frac{p_c}{p_1} - 1 \right) = \frac{2}{\gamma 2^2} (1.29 - 1) = 0.104$$

$$D = \int_0^{r_b} 2\pi r p_c dr - \int_0^{r_b} r 2\pi p_1 dr = \int_0^{r_b} 2\pi r (p_c - p_1) dr = \pi (p_c - p_1) r_b^2$$

oblique surface base
horizontal force

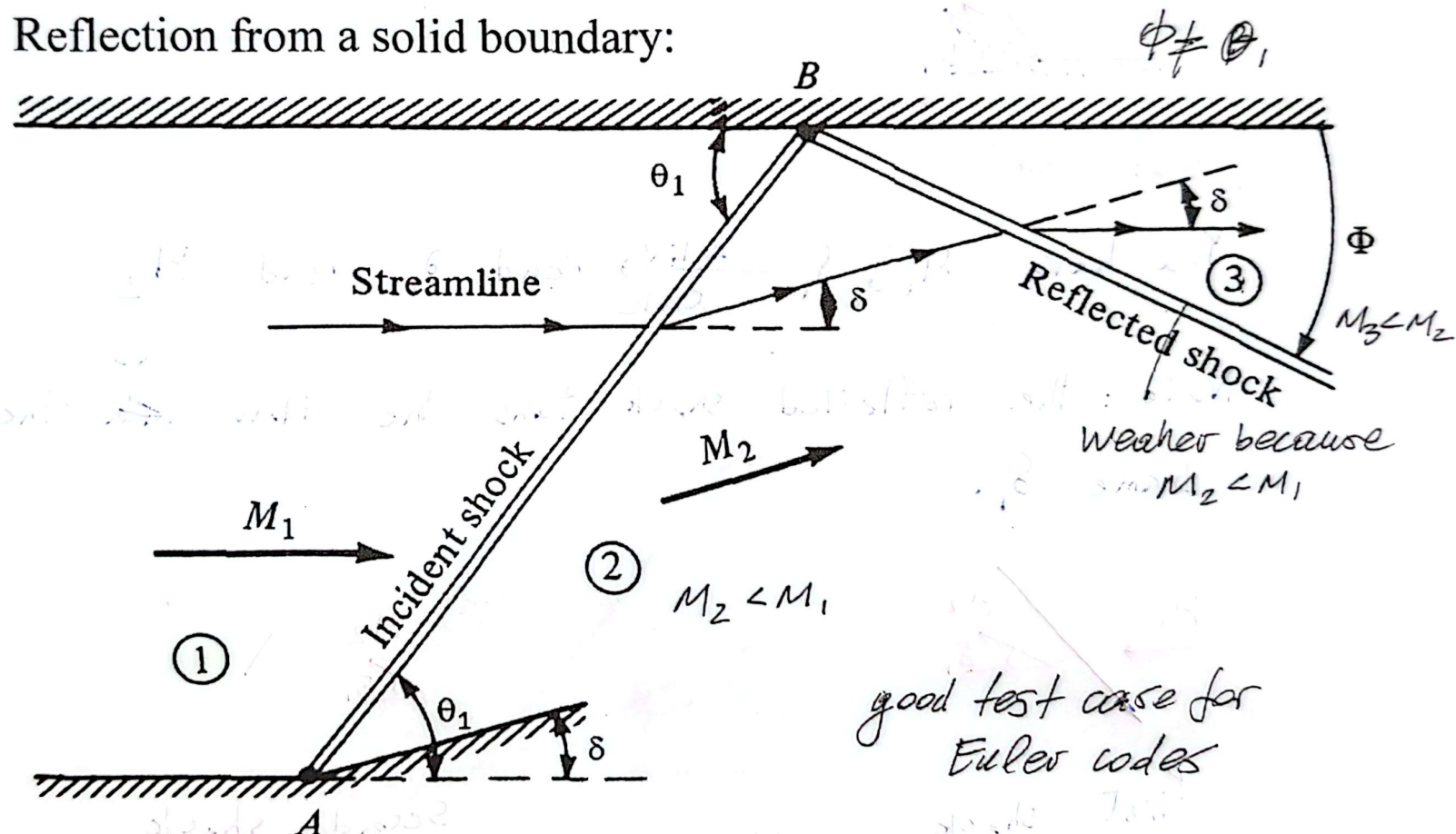
$$C_D = \frac{D}{q_1 \pi r_b^2} = \frac{\pi r_b^2 (p_c - p_1)}{\pi r_b^2 q_1} = \boxed{C_p} \quad (C_D = C_p)$$

= 0.104



C_D for cone is C_p

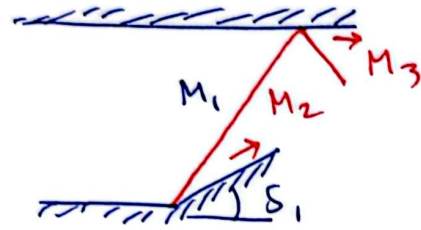
Reflection from a solid boundary:



2 oblique shocks sequentially; second shock is weaker $\phi \neq \theta_1$

(because it starts at a lower M)

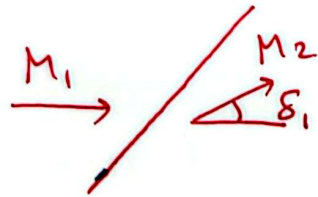
from $M_1, \delta \rightarrow \theta_1, M_2$; $M_2, \delta \rightarrow M_3, \theta_2$ $\phi = \theta_2 - \delta$



"red: shock"

You have $M_1, \delta_1 \xrightarrow[\text{calc}]{\text{online}}$ read θ_1 , read M_2

Note: the reflected shock turns the flow ^{as} ~~with~~ the same δ_1 .

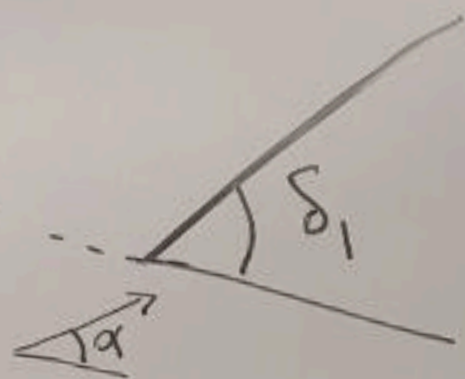


first shock

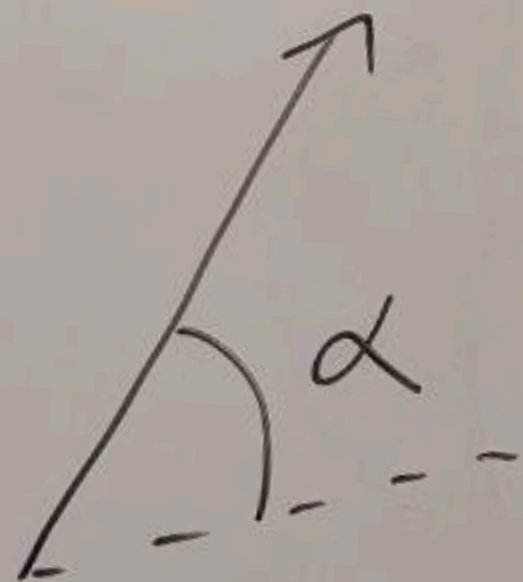
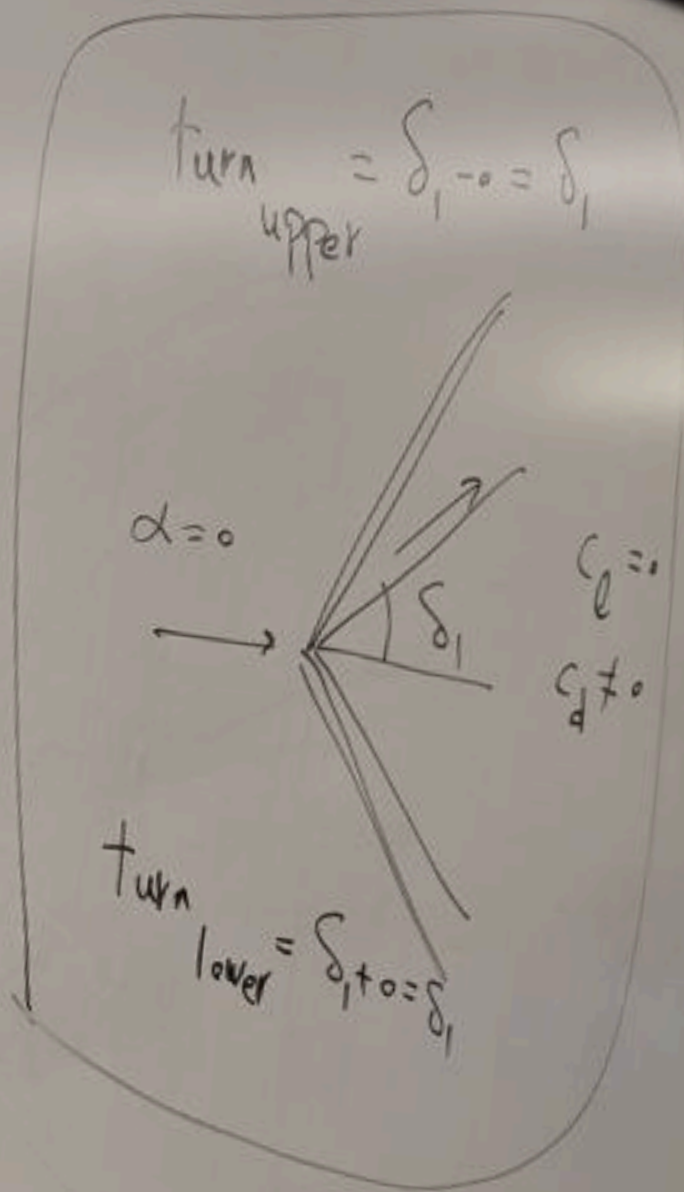
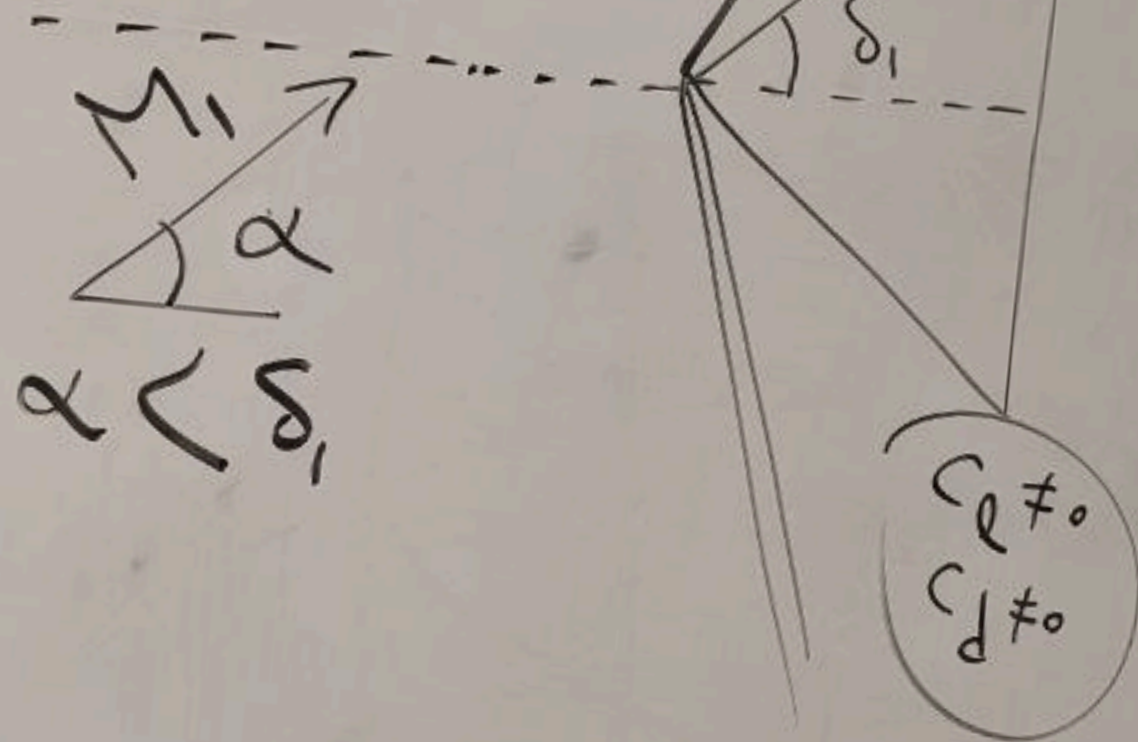


second shock

To solve second shock, use $M_2, \delta_2 = \delta_1 \xrightarrow[\text{calc}]{\text{online}}$ find M_3
and other data

upper side 

turn angle: $\delta_1 - \alpha$



turn angle: $\delta_1 + \alpha$